SUSTAINABILITY

Next-Generation Sustainability Dashboards: AI-Powered Optimization for Data Centers

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Agenda

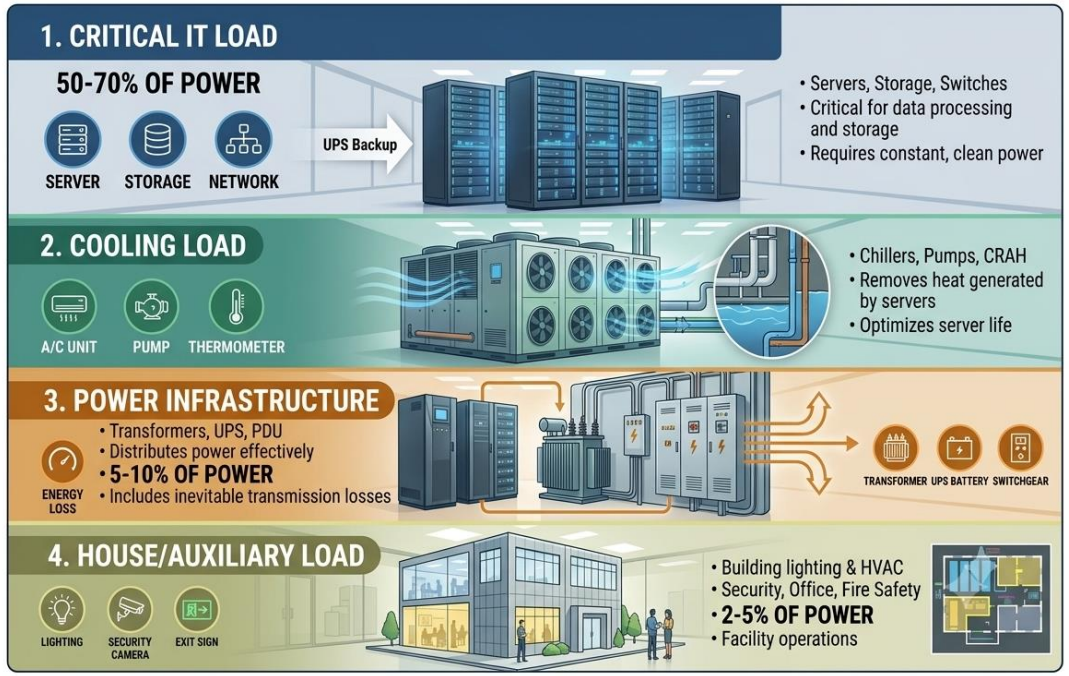
- Importance of Sustainability Dashboard
- Measuring and Reporting
  - KPIs, Matrices and Carbon Calculation
  - Sustainability Dashboard, Dashboard for Software
- Optimization
  - Time, Space Analysis
  - Carbon Capping
  - Workload placement and Traffic routing

Importance of Sustainability Dashboard

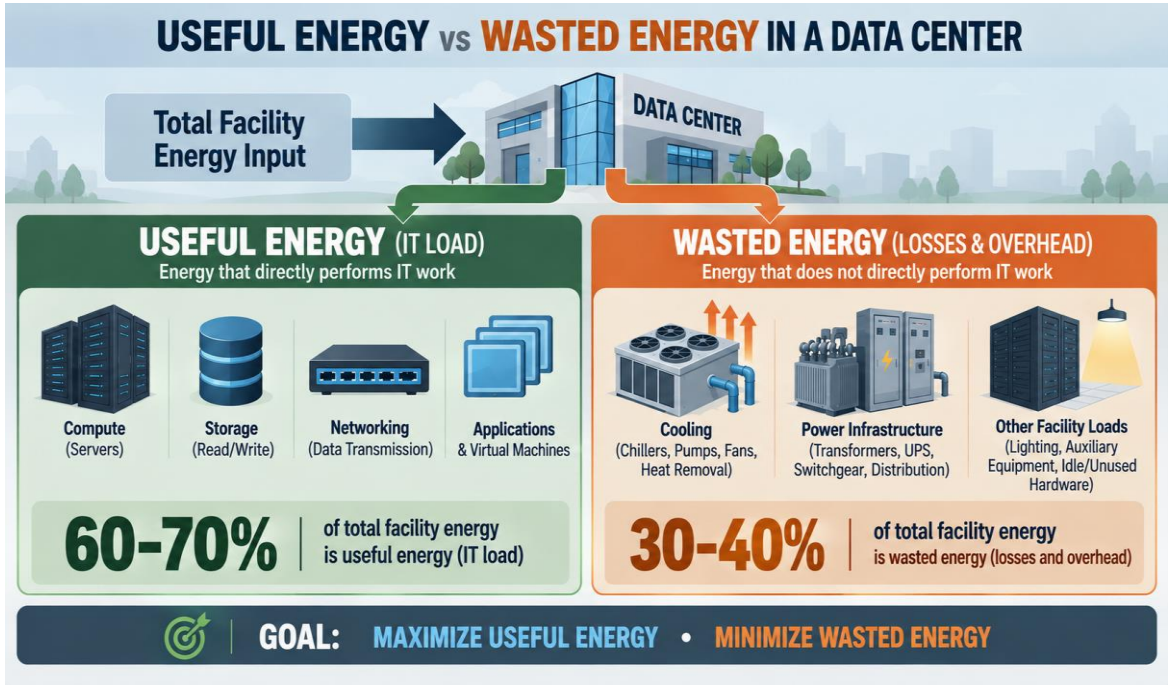
- Sustainability Goals
  - Environmental commitment
  - Carbon tax
- Measure and Report
  - Reporting helps understand the current usage and trend
- Optimize – Actions based on data
  - Act based on usage trends and pattern
  - Usage of Analytics and AI for actionable steps

Measuring and Reporting

Types of loads in Data center



Useful Energy and Wasted Energy



KPIs and Matrices

Key Metrics for Data Center Sustainability

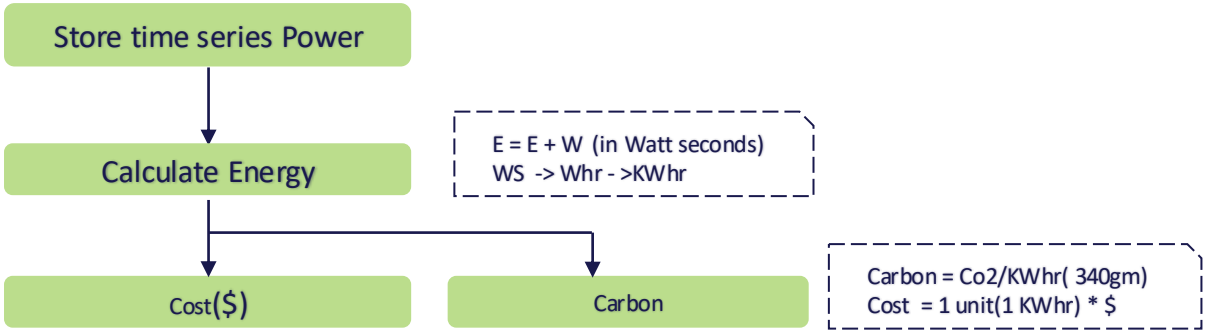
Metric	Definition / Focus
Energy Use	Total energy consumed, measured in kWh (e.g., monthly or yearly).
Carbon Emissions (Scope 1, 2, 3)	Total CO_2e emissions associated with energy use (PUE and grid emission intensity).
Renewable Energy Factor (REF)	Percentage of total energy from renewable sources (on-site or off-site).
Server Utilization Rate	Percentage of server resources (CPU/RAM) actively performing useful work (vs. idle or comatose).
Emission Factor	Coefficient used to convert energy consumption (kWh) into carbon emissions (kg CO_2e/kWh).
Product Carbon Footprint (PCF)	Lifecycle greenhouse gas emissions associated with a specific server, product or service (e.g., kg CO_2e).

Data Center Performance Indicators

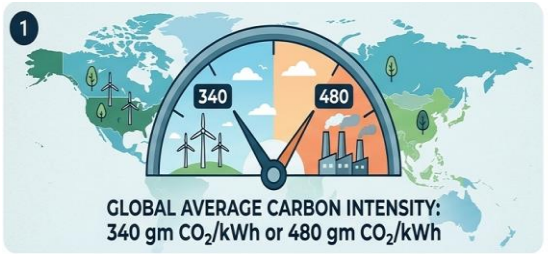
KEY METRIC	FORMULA / MEASUREMENT	FOCUS / PURPOSE	TARGET
PUE Power Usage Effectiveness	$\div$ Total Facility Energy / IT Equipment Energy	Measures overall facility energy overhead.	Ideal: **1.0** Typical target: **1.4**
WUE Water Usage Effectiveness	$\div$ Annual Water Use (liters) / IT Energy (kWh)	Tracks water consumed per unit of IT energy.	**Lower is better** Target depends on site and cooling design.
CUE Carbon Usage Effectiveness	$\div$ Total Data Center Carbon Emissions (kg CO_2e) / IT Energy (kWh)	Tracks carbon emissions per unit of IT energy.	**Goal: reduce operational emissions** toward net zero.

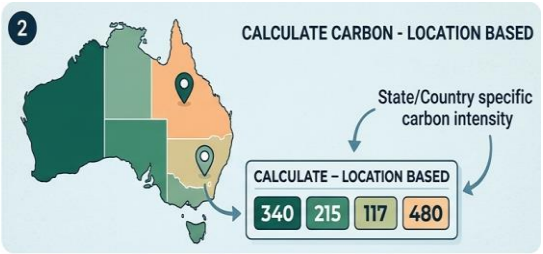
Energy and Carbon calculation

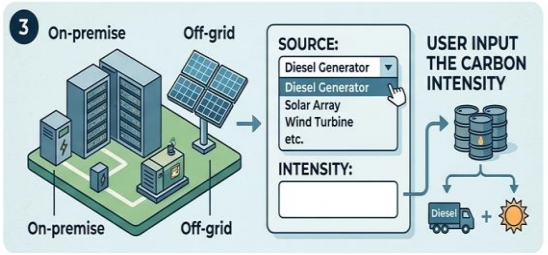
Calculation

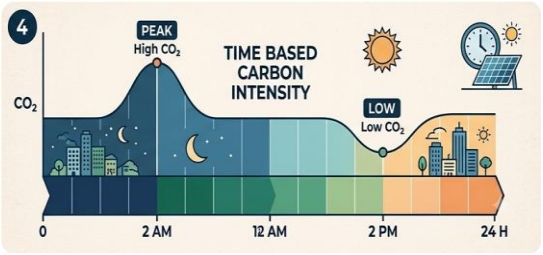


Approaches

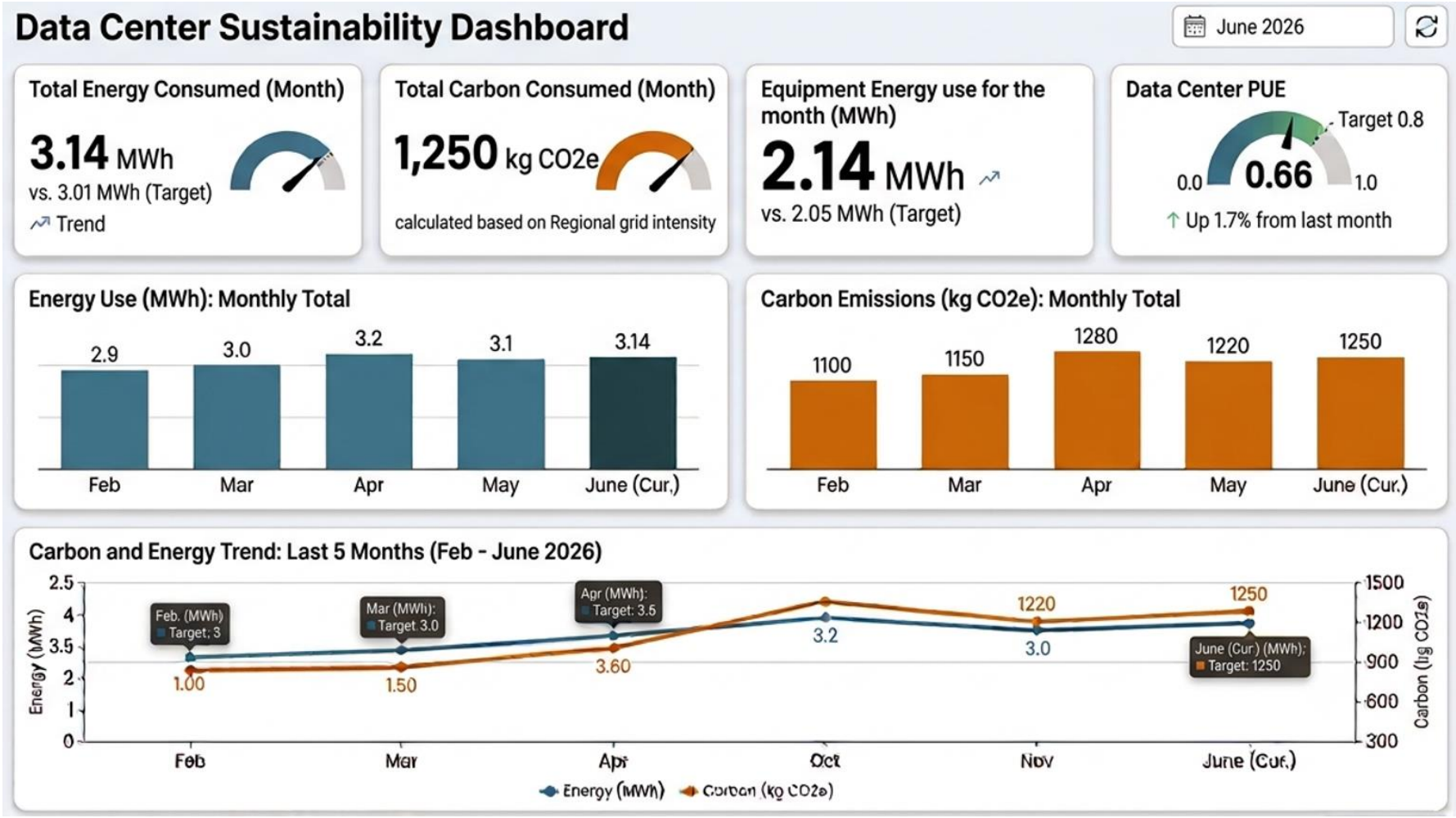








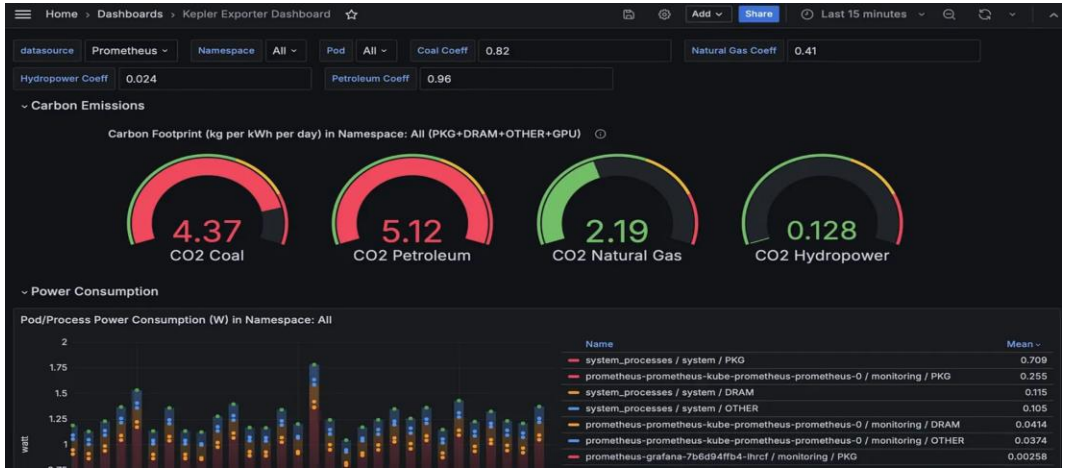
Dashboard


Data Center Sustainability Dashboard
June 2026
Total Energy Consumed (Month)
3.14 MWh
vs. 3.01 MWh (Target)
Trend
Total Carbon Consumed (Month)
1,250 kg CO2e
calculated based on Regional grid intensity
Equipment Energy use for the month (MWh)
2.14 MWh
vs. 2.05 MWh (Target)
Data Center PUE
Target 0.8
0.0
0.66
1.0
↑ Up 1.7% from last month
Energy Use (MWh): Monthly Total
2.9
3.0
3.2
3.1
3.14
Feb
Mar
Apr
May
June (Cur.)
Carbon Emissions (kg CO2e): Monthly Total
1100
1150
1280
1220
1250
Feb
Mar
Apr
May
June (Cur.)
Carbon and Energy Trend: Last 5 Months (Feb - June 2026)
Energy (MWh)
Carbon (kg CO2e)
Feb. (MWh): Target: 3
Mar (MWh): Target: 3.0
Apr (MWh): Target: 3.5
June (Cur.) (MWh): Target: 1250
1.00
1.50
3.60
3.2
3.0
1220
1250
Feb
Mar
Apr
Oct
Nov
June (Cur.)
Energy (MWh)
Carbon (kg CO2e)

Dashboard for Software

- Most of the tools uses RAPL to measure the power consumed at processor level. Some tools report it **process level** using **CPU usage time, Wakeup time** etc.
- Tools/Libraries
  1. CodeCarbon
  2. PowerTOP
  3. CO2.js
  4. Kanoppi
  5. Eco-CI
  6. Kepler
  7. kube-green

Kepler – K8S





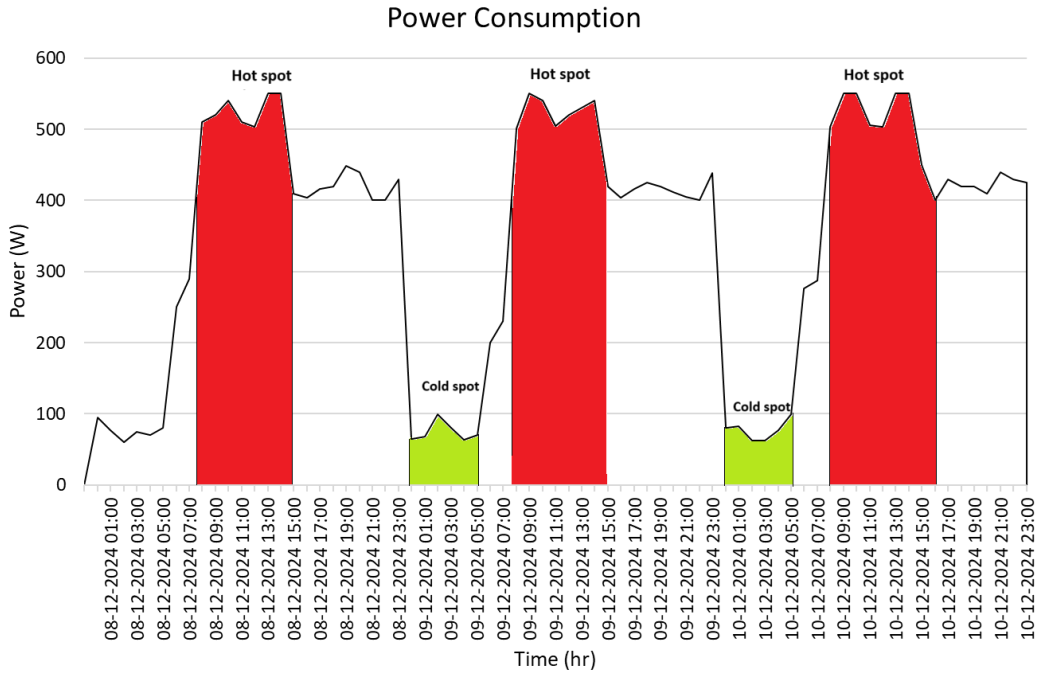
Optimization

Time Analyses

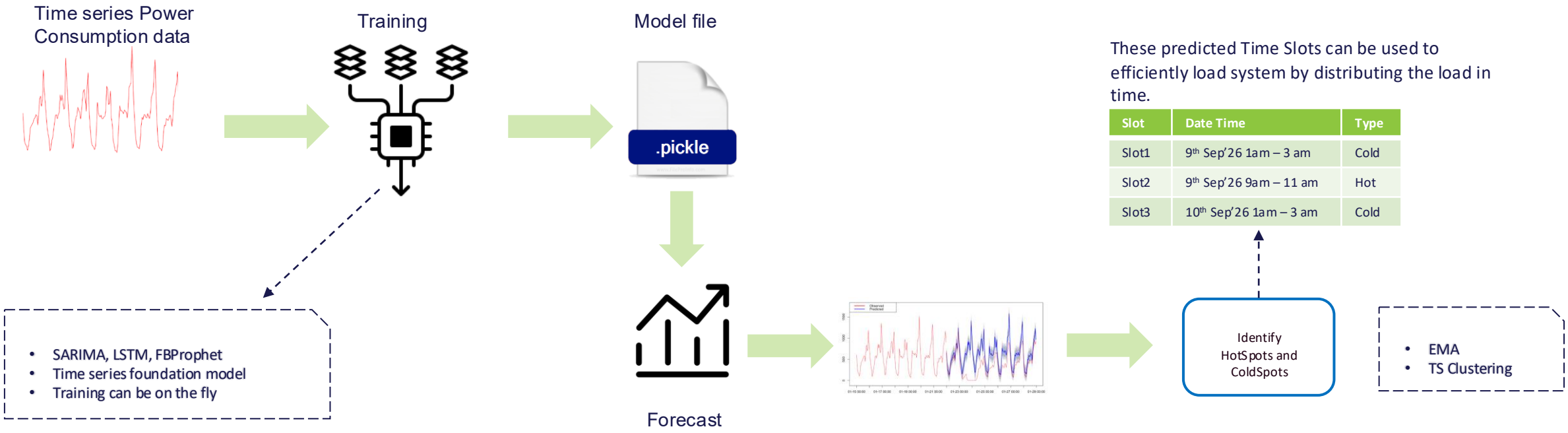
- Time-series energy data analysis is one of the most important enablers of data center power optimization because it transforms raw power measurements into actionable insights about energy consumption patterns, infrastructure efficiency, and operational risks.

Helps in

1. Demand Forecasting and Capacity Planning
2. Peak Power Reduction
3. Cooling Optimization
4. Anomaly Detection and Fault Prediction
5. Energy Efficiency Measurement
6. AI/ML-Based Power Optimization



Time Analyses – Forecasting Power



Feature Selection and Algorithms:

- Multiple features were looked at, like CPU, Mem Bus, I/O Bus Utilization
- Tried Mutli variant and Uni variant – Finalized Uni variant with Power Data

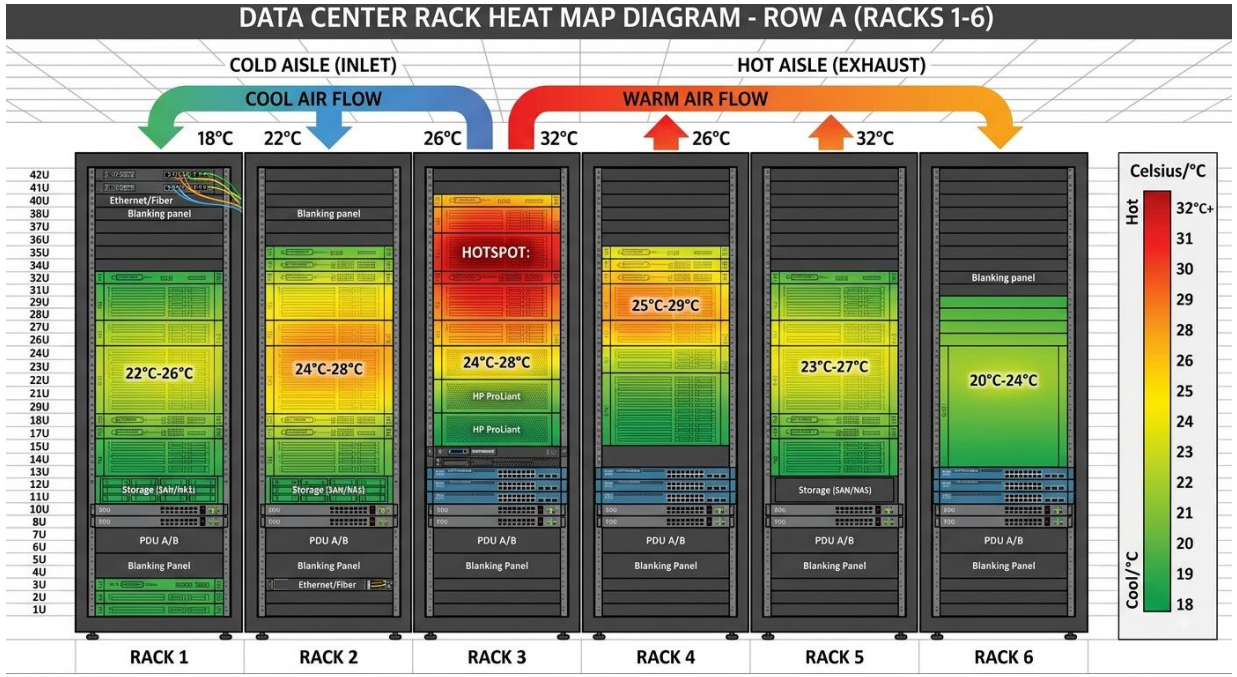
Model training

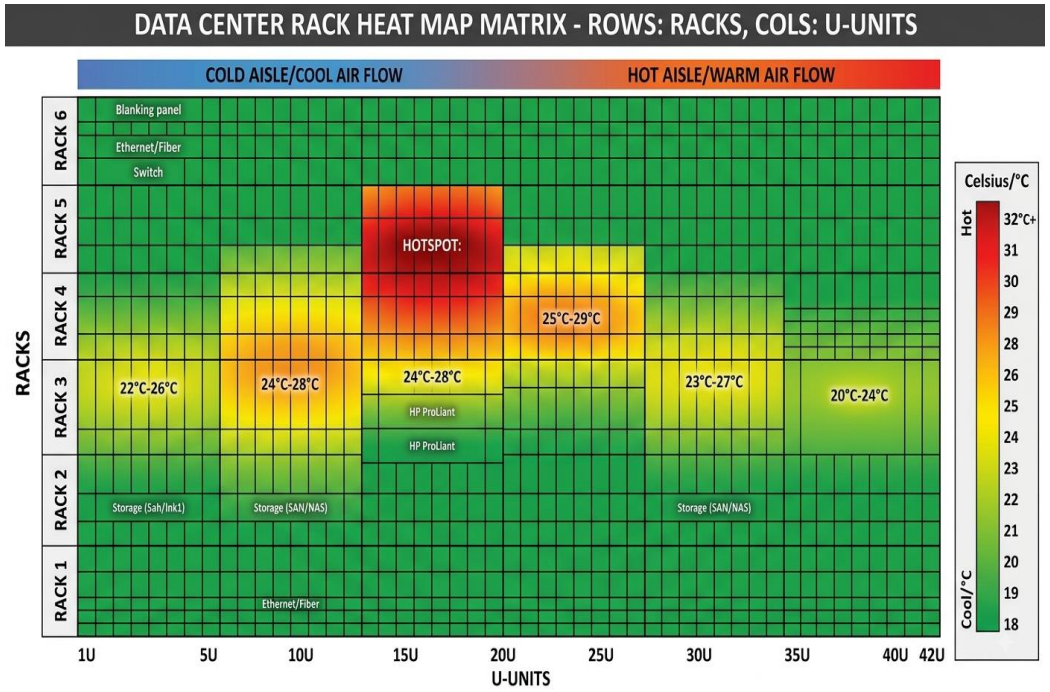
- Models tried SARIMA, LSTM, FBProphet – Finalized FBProphet
- Considered Training Time, Model Size and Inference Time
- Foundation models helps in Cold start

Determining Time Windows

- Few mathematical models were tried and Finalized EMA
- Initial Stage on trying TS Clustering

Space Analysis



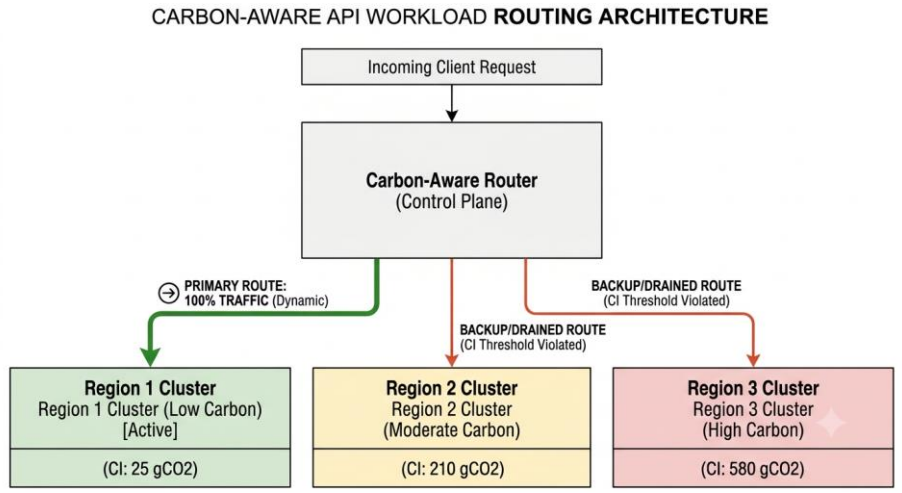


Workload Scheduling and routing – K8S

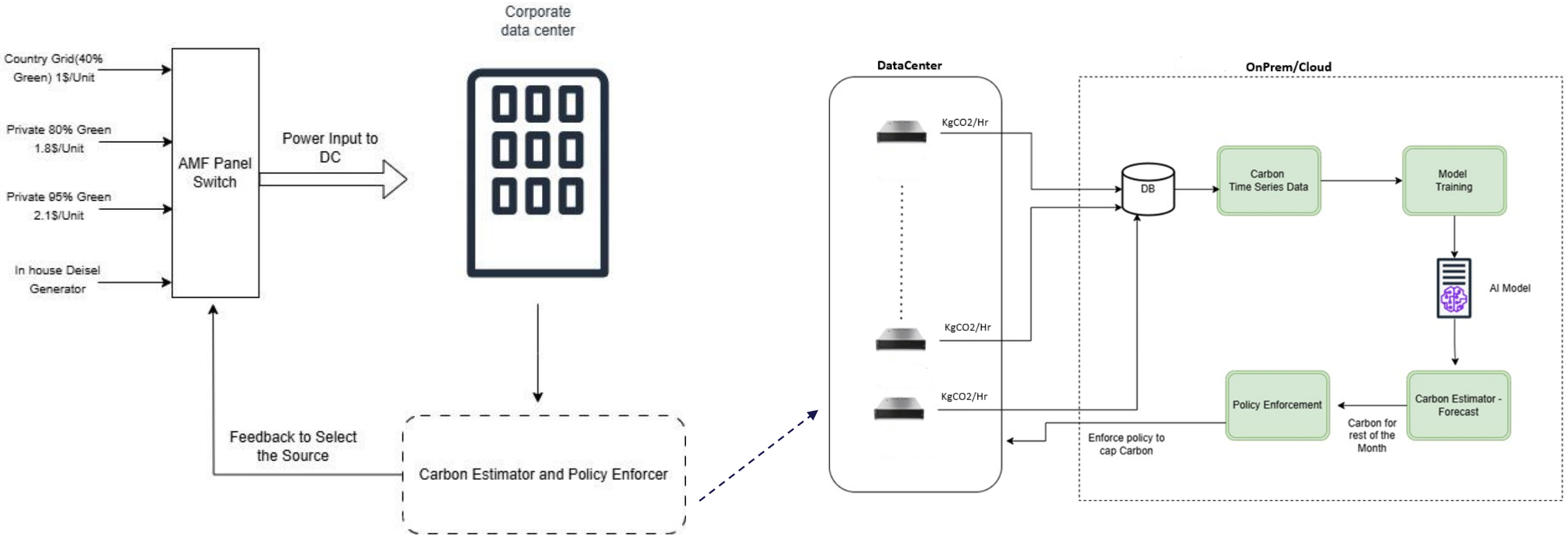
- Carbon Aware Workload placement
  - Location based
  - Time based – Carbon Intensive time slots
- Moving the Pods or Adjusting K8s scheduling policy based above
- Reporting the saved Carbon/Energy



- Routing the APIs based on carbon intensity of the location.
- Consider Time based carbon intensity in routing algorithm.



Carbon Capping



Carbon Capping

- Carbon capping and planning is super important considering the Carbon Taxes now enforced by governments.
- This is an expansion of power forecasting to plan the Carbon budgeting.
- Optimizing based on Sustainability goals and Carbon Tax
- Define Carbon budget or capping. Eg Monthly limit – E.g. 20KgCo2 per Rack
- Based on historical carbon, forecast for remaining carbon for the month.
- Enforce carbon usage by policy to individual server or group of servers
  - Tag servers that can be power capped, power down if under utilizes based on tag.
- If policy cannot meet the planned cap, feedback to switch panel to select the Greener Energy.

Call to Action

- Collaborate with the OCP community to define open sustainability standards for Sustainability related matrices
- Establish a workstream to facilitate discussions on Sustainability dashboards and Matrices
- References
  - Kepler documentation
    - https://sustainable-computing.io/kepler/developer/dev-docs/
  - Emission factor
    - https://www.epa.gov/climateleadership/ghg-emission-factors-hub
    - https://ourworldindata.org/greenhouse-gas-emissions

Thank You!

